Run an HMS API with AgentGate

Quick guide ? API setup, AgentGate test flow, and VS Code steps

What it does: AgentGate sends synthetic test inputs to a running API, maps its JSON response, checks expected behavior and quality limits, then creates a PASS/FAIL report. Smoke mode runs selected checks once and does not use an LLM judge.

Run from a terminal

1. Start the API from the repository root; wait for model loading. Keep this terminal open.

```
python -m uvicorn api:app --host 127.0.0.1 --port 5005
```

2. Configure AgentGate in `Agent_testing\agentgate`. Point it at the API endpoint and map request/response fields. Example for the API file legacy route:

```
type: http
http:
  url: http://127.0.0.1:5005/legacy/route-symptom
  method: POST
  request_template:
    query: "{{input}}"
  response_map:
    text: raw.query
    final_state: raw
```

3. Create matching test cases. Use synthetic data, not real patient information. Each case needs an ID, category, input, non-empty expectations, and `tier: smoke`. The bundled demo retail dataset is not a suitable HMS dataset.

4. Validate and run from `Agent_testing\agentgate` (adjust config and dataset paths to your files):

```
python -m agentgate validate --dataset dataset\golden_cases.json
python -m agentgate run --tier smoke
  --agent config\agent-hms.yaml
  --dataset dataset\golden_cases.json
```

Do it in VS Code

1. Open the repository folder in VS Code: **File ? Open Folder?**.

2. Open the integrated terminal with **Ctrl+`**. Use the terminal dropdown or split-terminal button to open two terminals.

3. In terminal 1, run the API command above and leave it running. In terminal 2, change directory: `cd .\Agent_testing\agentgate`.

4. In the Explorer, edit/create the AgentGate HTTP YAML and synthetic dataset. Save, validate, and run using the commands above. Review `reports\latest\report.pdf` and `results.json`.

5. Stop the API with **Ctrl+C** in terminal 1 when done.

What AgentGate does for each run

Select	Loads cases/rules; smoke selects only smoke-tier cases.
Send	Substitutes input into the request template and calls the configured HTTP endpoint.
Map	Maps JSON response fields into AgentGate?s normalized result.
Check	Evaluates case expectations plus must-never, error-rate, and latency limits.
Report	Writes a gate verdict and PDF/JSON results under the selected reports folder.

Observed in the previous API test: health returned OK, but both POST requests to `/legacy/route-symptom` returned HTTP 422 (the API reported `request` as a required query parameter). AgentGate reported SMOKE FAIL: 0/2 passed and 100% request errors. Fix/verify JSON request-body binding before treating a later run as a functional pass.

Previous report: `Agent_testing\agentgate\reports\hms-api-file\latest\report.pdf`. A healthy server does not by itself prove the tested endpoint works.